📑 Basic Information:

Experiment No: 07

Date:

Title: Write a Java Program to Demonstrate Inheritance, Method Overriding, and Dynamic Polymorphism by Calculating the Rate of Interest for RBI, SBI, ICICI, and PNB Banks.

🎯 Objective /Aim:

To write a Java program to implement bank functionality using a base class (RBI) and derived classes (SBI, ICICI, PNB, etc.) to calculate the rate of interest, and to demonstrate the concept of inheritance, method overriding, and dynamic polymorphism.

System Requirements:

Hardware:

-> Standard PC/Laptop

-> Intel Core i3/i5 processor or equivalent

-> Minimum 4 GB RAM

-> Minimum 500 MB free

Software:

->Operating System: Windows / Linux / macOS

-> Java Development Kit (JDK 17 or JDK 21)

-> Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code

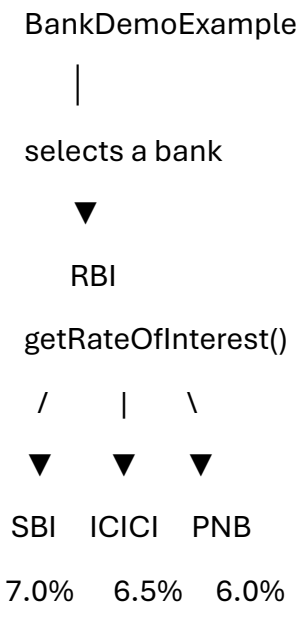
-> Command Prompt / Terminal

Algorithm/Logic Description:

1. Start the program.
2. Create a Customer class to store customer details such as name and customer ID.
3. Create an Account class to store account details such as account number and balance.
4. Create an RBI base class with a method getRateOfInterest() that returns an interest rate of 4%.
5. Create the SBI, ICICI, and PNB classes by inheriting the RBI class.

6. Override the getRateOfInterest() method in each derived class:
    - SBI returns 7%.
    - ICICI returns 6.5%.
    - PNB returns 6%.
7. Create the main class BankDemoExample.
8. Read the bank name from the user using the Scanner class.
9. Create an RBI reference variable.
10. Check the entered bank name:
    - If the bank name is RBI, create an RBI object.
    - If the bank name is SBI, create an SBI object.
    - If the bank name is ICICI, create an ICICI object.
    - If the bank name is PNB, create a PNB object.
    - Otherwise, display "Invalid Bank Name" and terminate the program.
11. Call the getRateOfInterest() method using the RBI reference variable.
12. Display the corresponding interest rate.
13. Close the Scanner object.
14. Stop the program.

UML Class Diagram:



Customer Account

- name - accountNumber

- customerId - balance

- displayCustomer() - displayAccount()

Source Code:

```
package record;
import java.util.Scanner;
// Stores customer details
class Customer {
  String name;
  int customerId;
  // Creates a customer with a name and ID
  Customer(String name, int customerId) {
    this.name = name;
    this.customerId = customerId;
  }
  // Prints the customer's details
  void displayCustomer() {
    System.out.println("Customer Name : " + name);
    System.out.println("Customer ID   : " + customerId);
  }
}
// Stores account details
class Account {
  int accountNumber;
  double balance;
  // Creates an account with an account number and balance
  Account(int accountNumber, double balance) {
    this.accountNumber = accountNumber;
```

```
this.balance = balance;
  }
  // Prints the account's details
  void displayAccount() {
    System.out.println("Account Number : " + accountNumber);
    System.out.println("Balance       : " + balance);
  }
}
// Base class that provides the RBI interest rate
class RBI {
  double getRateOfInterest() {
    return 4.0;
  }
}
// SBI inherits from RBI and overrides its interest-rate method
class SBI extends RBI {
  @Override
  double getRateOfInterest() {
    return 7.0;
  }
}
// ICICI inherits from RBI and overrides its interest-rate method
class ICICI extends RBI {
  @Override
  double getRateOfInterest() {
    return 6.5;
  }
}
// PNB inherits from RBI and overrides its interest-rate method
```

```
class PNB extends RBI {
    @Override
    double getRateOfInterest() {
        return 6.0;
    }
}
// Main class: asks the user for a bank and displays its interest rate
public class BankDemoExample {
    public static void main(String[] args) {
        // Scanner reads the bank name entered by the user
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the Bank name to find the rate of Interest: ");
        String bankName = sc.nextLine();
        // RBI reference can point to RBI or any of its child classes
        RBI r;
        // Select a bank object based on the user's input
        if (bankName.equalsIgnoreCase("RBI")) {
            r = new RBI();
        } else if (bankName.equalsIgnoreCase("SBI")) {
            r = new SBI();
        } else if (bankName.equalsIgnoreCase("ICICI")) {
            r = new ICICI();
        } else if (bankName.equalsIgnoreCase("PNB")) {
            r = new PNB();
        } else {
            // Stop if the entered bank name is not recognized
            System.out.println("Invalid Bank Name");
            sc.close();
            return;
```

```
        // The object's overridden method is chosen at runtime (dynamic polymorphism)
        System.out.println("RBI rate of interest is: " + r.getRateOfInterest() + "%");
        // Close the scanner after reading input
        sc.close();
    }
}
```

Relationship:

SBI —inherits—▶ RBI

ICICI —inherits—▶ RBI

PNB —inherits—▶ RBI

BankDemoExample —creates and uses—▶ RBI / SBI / ICICI / PNB

Output Screenshot:



Result:

The program demonstrates **inheritance** and **runtime polymorphism**: SBI, ICICI, and PNB inherit from RBI and provide their own interest rates. The program reads a bank name and displays that bank's rate. Customer and account details are defined but not used.